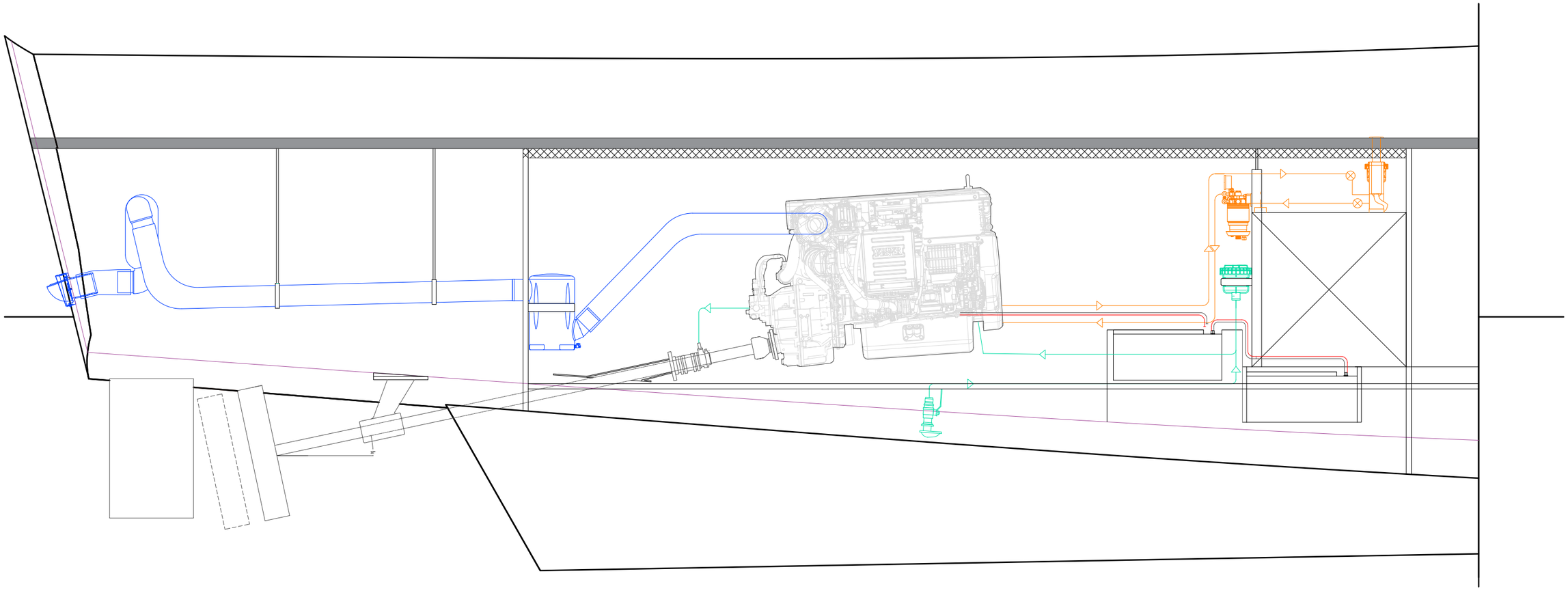
Shaft Line And Systems Installation Project

This project presents the design and integration of a complete propulsion system for a 45 ft composite cruiser, with a focus on system layout optimisation and overall installation coherence.

The vessel is powered by twin Volvo Penta D4 engines (230 hp each), paired with PRM750A3 gearboxes featuring a down-angle configuration, enabling a target maximum speed of 14 knots.

SOLENT UNIVERSITY SOUTHAMPTON

Profile View



Exhaust System

Fuel System

Cooling System

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	Drawn by: Roman Aime	Units: mm	Projection method: N/A
	Drawing number: 1		**Drawing title :**Profile View View Coaster_45_RA
	Edition: 1	Sheet: 2 of 4	

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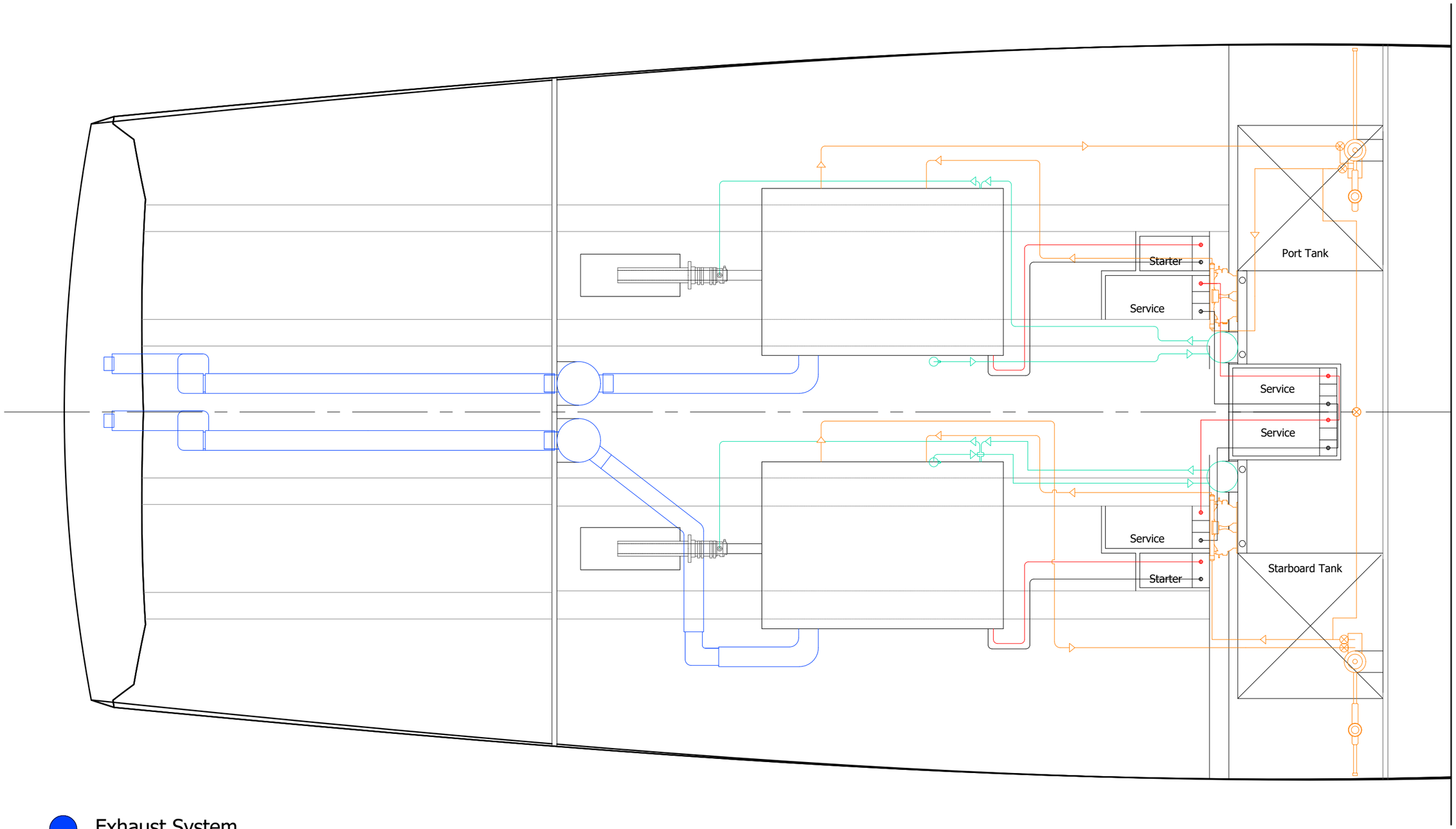
Top View



Exhaust System

Fuel System

Cooling System

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	Drawing number: 1		**Drawing title :**Top View View Coaster_45_RA
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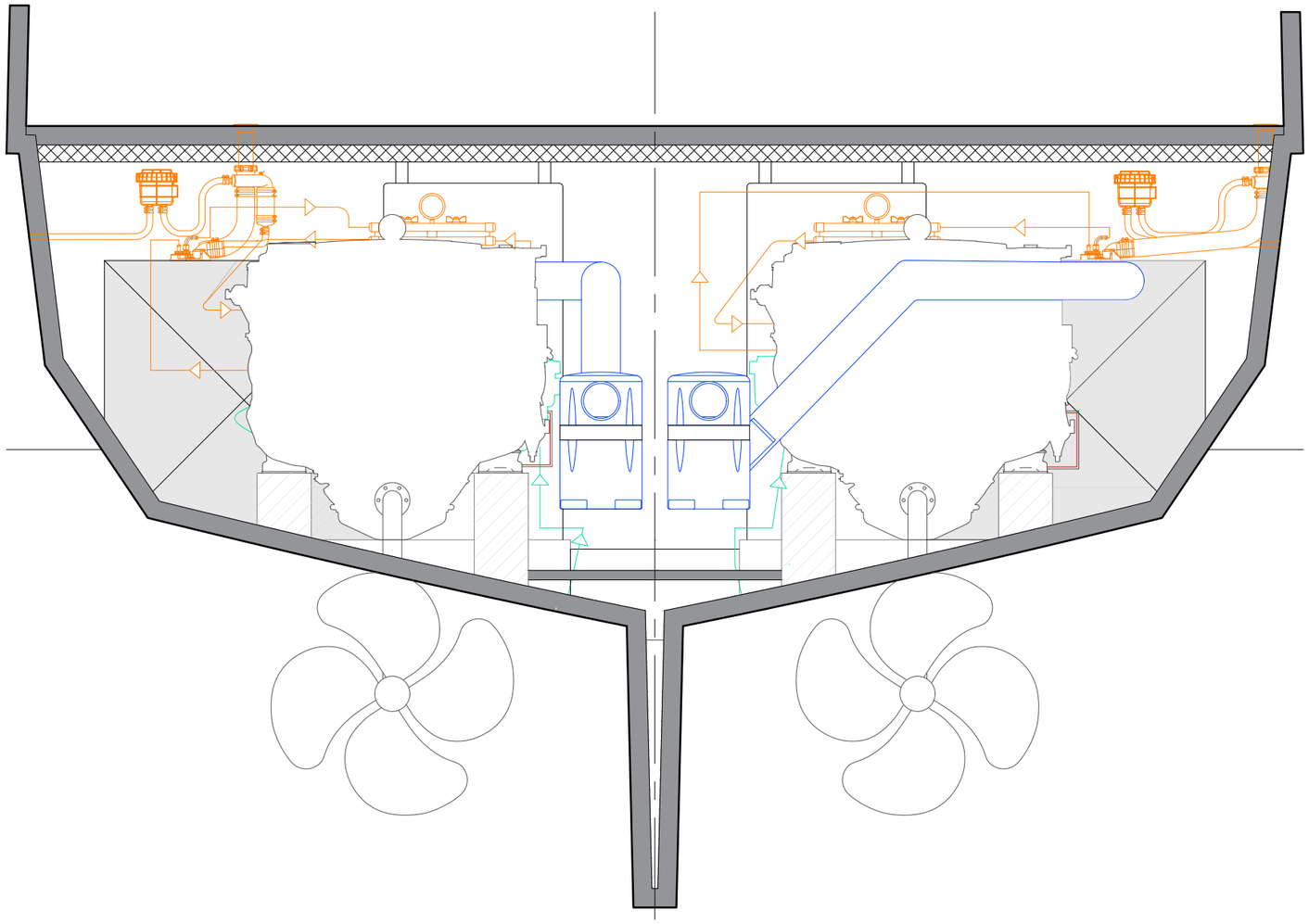
Aft Bulkhead View, 3.9m from the AP



Exhaust System

Fuel System

Cooling System

SOLENT UNIVERSITY	Issue date:28/11/2025	Scale: 1:15	File name: **Aft bulckead** Coaster_45_RA
	Drawn by: Roman Aime	Units: mm	Projection method: N/A
	Drawing number: 1		**Drawing title : Aft bulckead** Coaster_45_RA
	Edition: 1	Sheet: 3 of 4	

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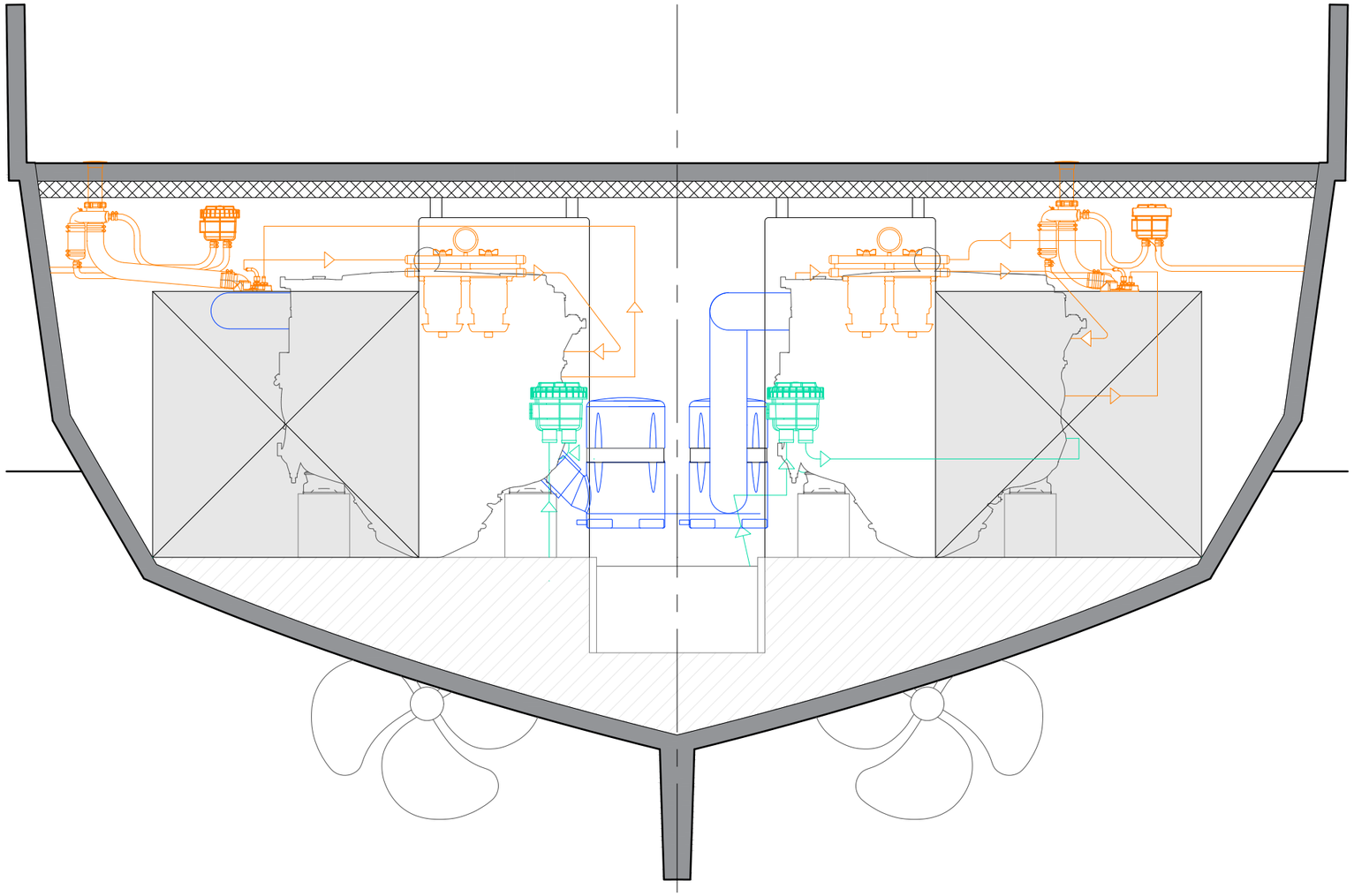
Saloon Bulkhead View, 6.7 m from the AP



Exhaust System

Fuel System

Cooling System

SOLENT UNIVERSITY	Issue date: 28/11/2025	Scale: 1:15	File name: Saloon Bulkhead ViewCoaster_45_RA
	Drawn by: Roman Aime	Units: mm	Projection method: N/A
	Drawing number: 1		**Drawing title :**Saloon Bulkhead View Coaster_45_RA
	Edition: 1	Sheet: 4 of 4	

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SHAFT LINE AND SYSTEMS INSTALLATION

SEF 536

Applied Marine Systems

Assessment Number : AE1

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1. Introduction :

The aim of this Report is to justify the technical choices for the installation of a preliminary propulsion system for a Coaster 45, a new composite cruiser boat, while optimising its layout. This boat is equipped with two Volvo Penta D4 engines, each with a power of 230 hp, coupled with a PRM750A3 gearbox with a down angle, which will allow it to reach the desired maximum speed of 14 knots.

2. Propulsion System :

2.1 Engine

The engine selected for the Coaster 45 is a Volvo Penta D4 from the leisure boat range, providing 230 hp. Its RPM of 3400 is lower than its competitors, which heavily influenced the choice and allowed for a pitch ratio of 0.65, keeping it within the cruiser boat range. Moreover, the D4 Volvo engine is widely used and known for its reliability and low fuel consumption relative to its power.

2.2 Gearbox (750A3 / 8° down angle)

This gearbox has been designed for use in pleasure craft. It is capable of operating at 4000 rpm continuously, with an 8-degree down angle. Being the only gearbox on the market with a ratio of 2.904 for 3400 rpm, this choice allowed us to achieve a pitch ratio within the cruiser range. Additionally, as demonstrated in the Excel sheet, the actual PSI remains well below the maximum PSI with a difference of 3.46. Therefore, the risk of cavitation is nil, confirming this choice.

2.3 Propeller (Vetus P4G 25x17 -VP4G25X17L)

This 4-blade propeller from Vetus has a DAR of 0.85, perfectly matching the calculated diameter using Gerr's propeller diameter displacement formula. This model was found at the Dutch distributor Boothelden.

2.4 Shaft (CJR, D = 50mm, Length =2400mm)

Using Rina's simpler formula, Gerr's formula, and Rina's complex formula, the final diameter chosen is 50 mm, which aligns perfectly with the chosen propeller that accepts this diameter. The selected material is Aquamet 22, due to its superior corrosion resistance, excellent toughness, and high mechanical strength. Finally, with

an added angle of 4° in addition to the 8° of the gearbox, the shaft measures 2400mm, which is very acceptable, reducing vibration and minimising the risk of misalignment.

2.5 Coupling (R&D Clamp Couplings)

The coupling chosen is an R&D Clamp Coupling with a 50mm Male Register. A flexible coupling would have been a more interesting option in case of misalignment, but the available space did not allow for this. As R&D is more compact, it was the final choice.

2.6 Shaft seal (RZWB50A Vetus)

The ZWB50A uses a double-lip seal which offers good sealing for the shaft while allowing slight flexibility to adapt to the shaft's small movements. An internal bearing helps absorb vibrations and reduce wear, ensuring optimal alignment. Additionally, the seal is lubricated and cooled by a constant water supply. Maintenance is also made easier thanks to its dripless functionality, minimising water leaks.

2.7 Sterntube (Eliche Radice for 50mm D.)

For the sterntube, the choice was made for the Eliche Radice for 50mm, which accommodates the 12° angle of the shaft. Moreover, the bronze material is durable, facilitating maintenance. A Vetus bearing is inserted to support the shaft.

2.8 P-Bracket (SGP PMWT050)

Finally, for the P-bracket, the choice fell on the manufacturer SGP with the PMWT050 model, which perfectly matches the calculations made in the Excel sheet, which specify a minimum length at the hull bottom of 200mm. The length being 260mm, this choice is confirmed.

3. Cooling system

3.1 Sea water cock (6016700 Sole)

The sea water cock from the brand Sole has a diameter of 38mm, and the complete unit consists of a water intake, filtering mesh, through-hull, ball valve, and ball valve. The filtering mesh must absolutely be installed facing the direction of navigation; otherwise, it could lead to overheating risks for the engine. The ball valve must always be below the flotation line.

3.2 Water Strainer (FTR330 Vetus)

This water strainer has an inlet and outlet diameter of 38mm, allowing a maximum capacity of 470l/min, which fits with the sea water cock's inlet. The sea water is then sent to the high-pressure pump of the D4 engine for the cooling of the fresh water circuit. The two water strainers are located at the entrance of the bilge for easy access.

4. Fuel System

4.1 Tank

The fuel system consists of two tanks with a capacity of 400L each, placed on either side and at the rear of the bilge to keep the centre of gravity as close as possible to the LCB of the Coaster 45. The two tanks are made of stainless steel for its resistance to corrosion, durability, and strength. Both fuel tanks are connected to each other.

4.2 Fuel Filter duplex (75340VTEB Vetus)

The choice of this duplex filter is a safety measure. One filter is active, while the other remains in standby. If there is an issue with one filter, it is possible to switch directly to the other. Additionally, it serves as both a water separator and fuel filter. This model supports up to 620L/h, which is ample for the Volvo D4 engines. The fuel is then drawn by the engine's fuel pump and directed towards the injectors. Finally, the fuel tanks are placed in the centre of the bilge to provide easy access.

5. Exhaust System

5.1 Waterlock / MGP102102 Vetus

The waterlock's function is to evacuate gas/water but also acts as a water trap, serving as a protective barrier for the engine. It is perfectly suited to the engine with a 102mm diameter hose. It is also rotatable at 360°, allowing for additional angles to enhance safety.

5.2 Gooseneck (LT102 Vetus)

The gooseneck serves to lift the hose above the DWL (45cm above), preventing water from flowing into the system. Between the gooseneck and the waterlock, the hose has a minimal angle, adding an extra obstacle to prevent water from flowing back.

6. Electrical System

The electrical system consists of four AGM 180Ah service batteries and one 120Ah battery for each engine, for a total of six batteries.

7. Insulation

The choice was made for foam PT245S Vetus Prometech Double Acoustic Insulation to reduce noise and vibration generated by the engines and mechanical systems.

8. LCG

After calculating the LCG with full and empty tanks, based on the AP, we obtained a difference of 0.025m. This is minimal and will not have a significant impact on the longitudinal trim.

Conclusion

In summary, the shaft line and system installation were designed with performance, easy access, and proper positioning of the components in mind.

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